Chen Xu

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EDUCATION

New York University, Tandon School of Engineering *Expected May 2029*

B.S. in Computer Science GPA: 4.0

- Core Member of the ICPC Team (Poly Programming Club)
- Member of the Tennis Club

COMPETITIVE PROGRAMMING

- Codeforces International Master. USACO Platinum. CSP-J National Champion (1st nationally)
- **ICPC Gold Medal:** 9th of 294 teams at the ICPC/CCPC Shandong Provincial Contest 2026. Ranked 4th among the 237 teams.
- **ICPC Gold Medal:** 38th of 418 teams at the 50th ICPC Asia Xi'an Regional, 2025.
- **ICPC Silver Medal:** 82nd of 280 official teams at the 49th ICPC Asia East Continent Final, 2024.
- **ICPC Silver Medal:** 60th of 324 teams at the 11th CCPC Chongqing Regional, 2025.

EXPERIENCE

Computer Vision Engineer at Dacheng Precision Equipment *Feb – Aug 2026*

- Sole developer of a CCD machine-vision system that measures battery-electrode segment widths in active production on **30+ machines** across 5 battery manufacturers, carried through 50+ iterative releases driven by real plant feedback
- Owned the full product: chessboard calibration, electrode measurement and QC, real-time TCP integration with the inspection line's control system.
- Built the edge-detection pipeline from primitives: banded |Sobel-x| gradient response → non-max-suppression multi-peak finding → per-peak RANSAC line fit (parameterized x = m·y + b so near-vertical edges stay stable), with brightness-step and angle filters to reject foil wrinkles and scratches as false edges
- Designed a binary-search algorithm that finds the minimal relaxation of three coupled detection gates (peak threshold, RANSAC confidence, angle tolerance) to recover the expected edge count while admitting the fewest false edges; calibrated against ground truth at $R^2 > 0.995$

PROJECTS

Usage-Aware GPU Memory Oversubscription — HAMi (open-source Kubernetes GPU virtualization) *2025*

- Diagnosed that HAMi oversubscribes VRAM against *requested* quota while ignoring its own per-container real-usage signal, a blind over-commit that triggers CUDA OOM when memory-tight jobs co-locate
- Closed the loop in the Go scheduler so admission becomes *real_used + request* $\leqslant$ *physical capacity*, safely reclaiming idle GPU memory;
- Validated trace-driven simulation and live on a real Kubernetes/V100 cluster: raised utilization 55%⟶78% (1.4×) vs. an optimally-tuned baseline and 2× (28%⟶56%, 2× the admitted pods) vs. the fixed-scaling production default

Agent Toolchain & LLM Infrastructure *2026*

- Built agent tooling on MCP. Authored custom MCP servers exposing tools to LLM agents, and drove multi-step coding workflows through agent harnesses including tool-use loops and subagent orchestration
- Hands-on with the LLM serving and agent stack: API integration, KV-cache behavior, prompt and context engineering, and computer-use agents (UI-TARS Desktop)

BookaCook — iOS Application (*$20K Hackathon Grand Prize*, 100+ downloads) *2025*

- Shipped full-stack iOS recipe platform; built Python web crawlers to scrape 1,200+ cookbooks and 1M+ recipes
- Integrated multi-model AI pipeline: TikTok API ingestion → Whisper for speech-to-text → GPT-4o for ingredient recognition from images → Llama 3 8B for structured recipe generation

TECHNICAL SKILLS

Languages: C++, Python, Go, Swift

ML & AI: agent harnesses, PyTorch, API integration, KV-cache, prompt & context engineering

Systems & Infra: Kubernetes, Git

Computer Vision: OpenCV, RANSAC line fitting

Mathematics: Probability, statistics, linear algebra, algorithms & data structures